

Muhammad Abdullah

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PROFESSIONAL SUMMARY

Information Technology graduate passionate about building intelligent, AI-driven systems through machine learning, computer vision, and data-centric development. Experienced in developing scalable applications with a strong focus on automation, data pipelines, and problem-solving. Actively exploring opportunities that blend AI, DevOps, and full-stack development, and scalable deployment.

TECHNICAL SKILLS

Programming: Python, OOP, SQL, Postgres

AI/ML Frameworks: OpenAI, PyTorch, TensorFlow, Keras, Scikit-learn, YOLOv9, OpenCV, Transformers, CNN, RNN, Fine-tuning, Hugging Face, LLM

Backend Development: Flask, FastAPI, Docker

Vector Databases & Search: RAG (Retrieval-Augmented Generation)

Data Processing & Tools: NumPy, Pandas, Matplotlib, Seaborn, WordCloud

DevOps & Other: AWS Cloud Services (EC2, S3, ECS, ECR, CloudFront), Docker, Kubernetes, CI/CD with Jenkins, Terraform (IaC), Linux Bash Scripting, Git, GitHub Actions

EXPERIENCE

Machine Learning Intern | Namal University (CAID)

- Worked on computer vision algorithms using OpenCV, YOLO, and PyTorch for face detection and recognition tasks.
- Collaborated with team to optimize models for deployment and improve detection accuracy.

T8LaunchPad Trainee | NASTP Lahore

- Intensive training in ML/Deep Learning, Full Stack Development, and DevOps with hands-on AWS services (S3, CloudFront, Route 53, EC2, Jenkins automation, Git integration, CI/CD pipelines, Docker, Kubernetes, Terraform, IaC, ECR, ECS, and GitHub Actions).

EDUCATION

University of Mianwali

BS Information Technology | CGPA: 3.01

Jul 2020 – Aug 2024

PROJECTS

ChatDoc – AI-Powered Document System (FYP) | Python, RAG, Vector Database, Flask, OpenAI

- Developed a Retrieval-Augmented Generation (RAG) system enabling users to query uploaded PDFs/books using natural language and receive context-aware responses.
- Built an end-to-end LLM pipeline using Hugging Face Transformers, embedding models, and vector indexing for semantic document retrieval.
- Integrated OpenAI APIs with a React frontend to deliver an interactive AI-powered document question–answering interface.

Real-Time Person Re-Identification Verification System | Python, PyTorch, OpenCV, DeepSORT, NumPy

- Developed a real-time person detection and re-identification system to prevent unauthorized boarding.
- Integrated YOLOv8 for person detection and used embedding-based techniques for passenger re-identification.
- Implemented a dynamic live count mechanism for accurate occupancy monitoring.

ChatBot – AI-Powered Food Allergy Management Chatbot | Transformers, Python, SQLite, Hugging Face

- Developed a transformer-based chatbot using DistilBERT for personalized food allergy guidance.
- Structured diverse datasets (CSV, PDF, text) into an SQLite database for efficient querying.
- Built an NLP pipeline for tokenization and fine-tuning achieving **87% response accuracy**.

Cloud-Native DevOps CI/CD Pipeline with AWS, Terraform & Docker | Terraform, Jenkins, GitHub Actions, Docker, Grafana

- Built an automated CI/CD pipeline using Jenkins and GitHub Actions to deploy a Dockerized Flask app to AWS ECS Fargate via ECR.
- Provisioned AWS infrastructure with Terraform (IaC) including VPC, EC2, S3, and IAM using reusable modules.
- Implemented PostgreSQL backups to S3 and monitoring with Prometheus, Grafana, and Node Exporter.

CORE COMPETENCIES

Machine Learning Engineering, Deep Learning Architecture Design, Computer Vision Systems, Python Development, AI Model Deployment Optimization, Full-Stack Development, Data Pipeline Design (CI/CD), Problem-Solving, Technical Collaboration & Leadership, AWS Services.